

price from P_w to P_a . This leads to additional terms of trade loss $e+f+g$. Thus, export subsidy negatively affects the national welfare.

Table 6.1 Welfare Effects of an Export Subsidy

	Exporting Country
Consumer Surplus	$-(a + b)$
Producer Surplus	$+(a + b + c)$
Govt. Revenue	$-(b + c + d + e + f + g)$
National Welfare	$-(b + d + e + f + g)$

6.4 NON-TARIFF BARRIERS

Besides tariff barriers, various non-tariff barriers are also prevalent in various countries. Just like tariff barriers, non-tariff barriers also influence the global economy by limiting free trade. Various non-tariff barriers include quantitative restrictions like import quotas, Voluntary export restraints, Exchange controls, Import deposit schemes, Health and safety standards, Customs valuation procedure, Local content requirements etc. This section will discuss these non-tariff barriers and their impact on the economy.

6.4.1 Import Quota

Import quotas impose direct restrictions on the quantum of imports into a country.

In practice, quotas are administered through a system of import licenses. Only license holders can import specified quantities of the imported goods into the domestic market. When imports are limited, domestic demand for the good exceeds domestic supply plus imports.

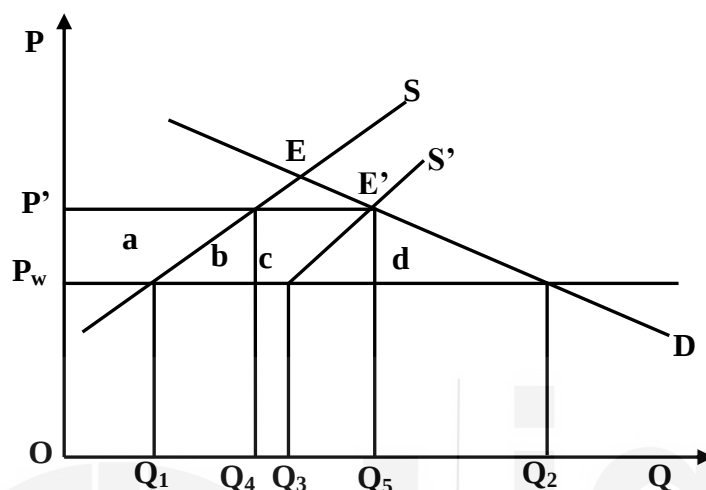
Thus, you will see that with a quota the domestic price of an imported good will always be higher than its world market price. License holders buy the imported goods at world market prices and then sell at higher prices in the domestic market. Import quota raises the domestic prices by the same amount as a tariff that limits the imports to the same level except when it is a monopoly in the domestic market.

In what follows we will examine the impact of an import quota under different market structures in the domestic economy. We first discuss the case of perfectly competitive markets and then that of monopoly.

1) Import Quotas with Perfect Competition

Figure 6.5 demonstrates the effect of an import quota when markets are perfectly competitive. D and S represent the demand and supply curves for the good before

quota imposition. Under free trade, the world price P_w prevails, total domestic production is Q_1 , demand is Q_2 , and $Q_2 - Q_1$ amount is imported. Now suppose an import quota is imposed, which restricts imports to $Q_3 - Q_1$ (where, $Q_3 - Q_1 < Q_2 - Q_1$). Immediately with quota imposition, at the world price P_w , domestic demand falls short of total domestic production plus imports. This excess domestic demand drives up prices in the domestic market, till the market clears.



Source: Figure 6.5 in Sikdar (2003)

Figure 6.6: Effect of an import quota when markets are perfectly competitive

The quota effectively shifts the domestic supply curve to S' , by the amount of the quota. The economy moves to the new equilibrium E' , where the price has risen from P_w to P' , domestic production has increased from Q_1 to Q_4 while domestic demand has fallen from Q_2 to Q_5 . At E' , imports, restricted by the quota, are equal to the amount $Q_4 - Q_5$ (note that $Q_3 - Q_1 = Q_5 - Q_4 = \text{import quota}$). A tariff rate equal to $P' - P_w$, is the tariff equivalent of the quota. It would have restricted imports to the same level as the quota and had the same effect on domestic prices. However, implementing the tariff equivalent of a quota may not always be feasible, as the rate may be too high to be acceptable. You should see that in the case of a tariff, a quota involves a loss in consumer surplus equal to the area $(a + b + c + d)$. This is offset by a rise in producer surplus equal to the area a . But an important difference between tariffs and quotas arises from the fact that with a quota the government does not earn revenues as in the case of a tariff. The area c , therefore, does not accrue to the government, rather it represents the quota rent, which may be captured by the import-license holders, who buy at the world price P_w and sell at a higher price P' , making a profit of $(P' - P_w)$ per unit of imports. If c accrues to the license holders and is counted as part of the social gain, then the social loss from the quota is equal to the area $(b + d)$, the same as in the case of a tariff.

Often foreign exporters have the right to sell directly in the domestic market. In that case, the quota rent c would accrue to foreigners and it would be a social loss from the domestic country's point of view.

Another disturbing possibility, and one that is often observed in practice, is that the quota rent may not accrue to license holders. Rather it may be dissipated in rent-seeking activities, like paying bribes to acquire import licenses. In that case, area c would be a social loss and the total cost imposed by the quota would equal the area $(b+c+d)$, which is more than in the case of an equivalent tariff.

Governments in developing countries have the option to auction import licenses. A competitive bidding process would drive the price of licenses up to $(P' - P_w)$ per unit of imports and the government would earn revenue equal to the area c . If this process worked smoothly, the effects of a tariff and quota would be equivalent. However, sometimes, the auctions may not be competitive and collusion among bidders might subvert the entire process.

2) Import Quotas with Monopoly

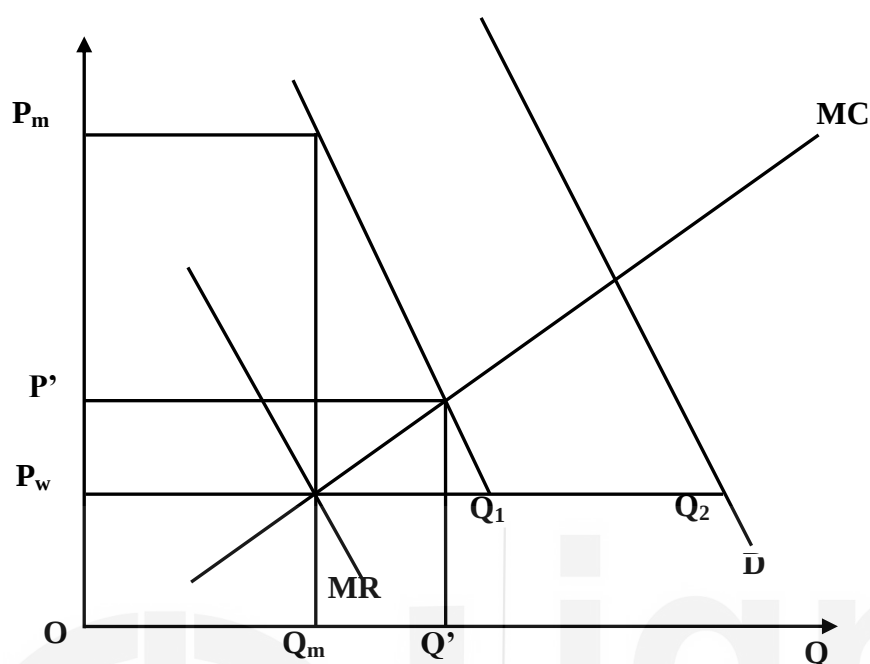
When there is a domestic monopoly, an import quota leads to a greater loss in social welfare, as compared to perfect competition. We will see that in the presence of a monopoly, an import tariff should be preferred to a quota from the efficiency and social welfare point of view.

When there is a tariff t , imports are freely available at a price $(P_w + t)$. So the monopolist cannot charge a higher price than this level, for if he did domestic consumers would go for imports and his sales would be reduced to zero. Thus, a tariff effectively imposes a price ceiling. However, with a quota, the monopolist can charge a price higher than the import price. In this case, he will not lose his entire market share, since imports cannot exceed the limit set by the quota. Thus a quota preserves the monopolist's price-setting power to a large extent.

This argument is illustrated in Figure 6.6. Under free trade, D is the demand curve facing the monopolist and the domestic price of the good is the world market price, P_w . Suppose, an import quota is set, limiting imports to the amount Q_1Q_2 . With the quota, the demand curve facing the monopolist shifts inwards by the amount of the quota, at all prices above P_w , because imports have reduced the monopolist's market by Q_1Q_2 . Post quota imposition, the relevant marginal revenue curve facing the monopolist is MR , corresponding to the new demand curve.

In this situation, the monopolist will maximise profits by producing an output of Q_m , at which marginal revenue equals marginal cost. This output will be sold at the price P_m , read off the demand curve. Now $(P_m - P_w)$ is the tariff equivalent of the quota or the quota rent earned by the import license holders. Note that with perfect competition in the domestic market, the MC curve would have been the supply curve, and the market price would have been P' and output, Q' . Clearly with a monopoly the outcome is more inefficient, compared to perfect competition. The monopoly output is lower ($Q_m < Q'$) and the price is higher ($P_m > P'$), leading to greater welfare losses. From this analysis it should be clear to you why tariffs are preferred to quotas, especially when the domestic producer

wields monopoly power. GATT negotiations have tried to phase out quantitative restrictions and replace quotas with tariffs.



Source: Figure 6.5 in Sikdar (2003)

Figure 6.5: Effect of an import quota when the market is a monopoly

6.4.2 Voluntary Export Restraint

A variant of the import quota is the voluntary export restraint (VER), also known as a voluntary restraint agreement (VRA). A VER is a quota on trade imposed from the exporting country's side instead of the importers. This is a self-imposed restriction on the quantity of a good permitted to export to another country.

VERs are export quotas negotiated by an importing country with its trading partners. Under this, importing countries request other countries to voluntarily lower their exports to protect domestic businesses. Under the negotiation, exporting country voluntarily agrees to restrict the quantity of exports. VERs had been a popular way for importing countries to protect their domestic industries without going into a trade war with exporting countries.

Historically, it was used largely by developed countries on textiles, footwear, steel, automobiles, etc to impose their own restrictions rather than facing tariffs and quotas. It was one of the popular measures of protectionism during the 1980s. But the WTO members agreed to not use any new VERs in 1994. As per many economists, Voluntary Export Restraints are ineffective over a longer period. For instance, The U.S. automobile companies were facing more competition from efficient Japanese automobiles so Japan imposed Voluntary Export Restraints (VERs) on the exports of cars manufactured in Japan to the U.S. but this non-tariff measure proved to be ineffective.

In Uruguay's negotiation round, World Trade Organization (WTO) does not prevent governments from imposing VERs on exporters through negotiation. However, the governments of the importing and exporting countries are sometimes able to convince exports to put VERs on the quantity of the exports. A VER is always more costly to the importing country than a tariff that limits the imports by the same amount.

Table 6.2 Summary of Impact of Tariff and quantitative restrictions on the Economy

	Tariff	Export subsidy	Import Quota	Voluntary export restraint
Producer Surplus	Increases	Increases	Increases	Increases
Consumer surplus	Falls	Falls	Falls	Falls
Government revenue	Increases	Falls (government spending rises)	No change (rent to license holders)	No change (rent to foreigners)
Overall national welfare	Ambiguous (falls for a small country)	Falls	Ambiguous (falls for a small country)	Falls

Source: Krugman, P. R., & Obstfeld, M. (1997). *International economics: Theory and policy*. Addison Wesley.

6.4.3 Other Non-Tariff Barriers

The following are some of the other important NTBs commonly used by countries following protectionist policies:

1) Exchange controls:

Exchange controls are restrictions imposed by countries' Central Banks that directly limit domestic residents' ability to acquire foreign currency in exchange for domestic currency. For instance, one method of imposing exchange controls involved acquiring the Central Bank's permission to hold foreign currency bank accounts.

2) Import deposit schemes:

These are rules imposed by countries' Central Banks, which tend to restrict imports by making them more expensive. For instance, under the rules importers are required to deposit a certain amount (usually in proportion to the value of the imported good) with the Central Bank, which effectively raises the cost of importing.

3) Health and safety standards:

Often importing countries insist that imported goods meet certain minimum health, safety and environmental standards. Meeting the standards would raise costs for the exporting country. Presumably, the underlying motive for standard imposition is to safeguard the health and general welfare of domestic residents of the importing nation. However, in practice, these standards are often used by developed nations to restrict imports originating from low wage developing countries. For instance, faced with cheap manufacturing imports from low-wage countries like Malaysia, Indonesia and Thailand, developed countries now insist that these countries comply with certain minimum labour and environmental standards, which would raise their production costs.

4) Customs valuation procedure:

Under this procedure, the importing country would artificially enhance the value of the imported goods under some pretext, which would raise its duty under a system of ad valorem tariffs. For instance, Sikdar (2003, p.141) cites the example of the US valuing certain chemical imports at the 'American selling price' (rather than at the 'invoice' price or the 'world market' price) in the post-war period. However, this practice was discontinued since the Tokyo Round of negotiations.

5) Local content requirements:

This is a practice followed especially in developing countries, which requires that some stipulated portion of a final good be produced domestically. Or it may be stated that a certain specified fraction of the final goods price must represent domestic value added. The underlying logic is to promote the local production of certain intermediate goods. From the importing firm's viewpoint, there is no restriction on imports. For they can import more, if they also buy more from local firms. From the domestic intermediate industries' point of view, a local content requirement provides trade protection in the same way as an import quota.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) What are the effects of an import quota, when there is perfect competition in the domestic market??

.....

.....

.....

.....

- 2) What are the effects of an import quota, when there is a monopoly in the domestic market?

.....
.....

- 3) List various non-tariff barriers that countries use to restrict trade flows.

.....
.....
.....

- 4) What do you mean by Voluntary Export restraints?

.....
.....
.....

6.5 LET US SUM UP

After understanding the theories of protectionism in the previous unit, we discussed various instruments countries use to protect their domestic industries from free trade and international competition. We discussed tariffs and various non-tariff barriers to trade in this unit.

In this unit, we discussed various categories of a tariff like import tariffs, export tariffs, Specific tariffs, Ad Valorem tariffs, Compound tariffs, revenue tariffs, protective tariffs, Single-Column tariffs, Double-Column tariffs and Triple-Column tariffs. We also discussed the effective rate of protection, which measures the extent of protection granted to the domestic industry by the given structure of nominal tariff rates. Tariffs on raw materials and intermediates would raise the cost of production for domestic industry and will affect the effective rate of protection vis-à-vis nominate rate of tariff.

We also discussed the economic impact of Import tariffs in partial equilibrium analysis on various economic agents like consumers, producers and the government. We found out that while tariff increases producer surplus and government revenue, it hurts consumer surplus. We also examined the impact of the tariff on the Volume of trade and Terms of trade (TOT) in general equilibrium analysis in the case of a small country and a large country. Overall the national welfare falls for the small country.

We also discussed Export subsidies as government policies that are implemented to encourage the export of goods by incentivizing local producers through easy and cheaper loans, direct payments, tax benefits etc. We analysed its impact on the economy and compared it with the effect of the tariff on the economy. We analysed that while export subsidy is beneficial for the producer, it hurts

Consumer surplus and government revenue. Overall, export subsidies negatively affect the overall national welfare.

We also discussed quantitative restrictions like import quota and Voluntary export restraints and their impact on the economy. We identified that import quota and VER increase producer surplus and reduce consumer surplus. We also discussed about other non-tariff barriers like exchange controls, import deposit schemes, health and safety standards, customs valuation procedure and local content requirements

We also identified that tariffs lead to more efficient outcomes compared to non-tariff barriers like quotas, as in addition to restricting the flows of imported goods; tariffs also impose a price discipline on domestic firms. Further, while the government directly benefits from tariff revenues, the quota rent may be completely dissipated in wasteful rent-seeking activities, leading to a social loss.

6.6 KEY WORDS

Consumer Surplus

The gain to consumers when the price they are willing to pay for a given quantity of a good exceeds the amount they have to pay for that quantity.

Effective Rate of protection (ERP) It measures the extent of protection granted to the domestic industry by the given structure of nominal tariff rates.

Free Trade

When countries engage in international trade without imposing any policy restrictions on the free flow of goods and services across international boundaries.

Non-tariff barriers

A non-tariff barrier is any measure other than a customs tariff, that acts as a barrier to international trade.

Perfect Competition

Perfect Competition is a market condition characterised by perfect information among buyers and sellers. It consists of a large number of firms that are price takers and who sell a homogeneous product to a large number of buyers.

Producers Surplus

The gain to producers arises from the difference between the minimum price at which they are willing to supply a given quantity of a good and the price they receive by selling that quantity in the market.

Quota	A quantitative restriction on the quantum of imports is permissible that is often administered via the distribution of import licenses.
Tariff	A tax imposed on imported goods raises the price of such goods in the domestic market.
Terms of Trade	The ratio of the price received for a country's exports to the price paid for its imports.
Voluntary export restraint	It is a self-imposed trade restriction at the request of the importing country whereby an exporting country limits the number of goods of a particular nature that it can export to a specific country or region.

6.7 SOME USEFUL REFERENCES

Acharyya, Rajat (2014). *International Economics: An introduction to theory and policy*. Oxford University Press. Chapters 9 and 10.

Cherunilam, Francis (1997). *International Economics*. Tata McGraw-Hill Publishing Company Ltd. Chapter 10.

Krugman, P. and M. Obstfeld (2000). *International Economics: Theory and Policy*, Addison Wesley Longman: Singapore. Chapters 8, 9 and 10.

Salvatore, D. (2004). *International Economics*, Eighth Edition, John Wiley & Sons: Singapore. Chapter 8.

Sikdar, S. (2003). *Contemporary Issues in Globalisation: An Introduction to Theory and Policy in India*, Oxford University Press: New Delhi. Chapter 6.

6.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check your progress 1

- 1) Case I: Equal, Case II: Higher. Case III: Lower
- 2) The Single-Column Tariff also known as the uni-linear tariff system provides uniform tariff rates to all the commodities irrespective of the country of origin while in the Double-Column tariff system, different tariff rates are charged for different countries.
- 3) Refer to section 6.2.1

Check your progress 2

- 1) Refer to section 6.2.3
- 2) Refer to section 6.2.4
- 3) Refer to section 6.2.4

Check your progress 3

- 1) Refer to section 6.4.1
- 2) Refer to section 6.4.1
- 3) Refer to section 6.4.3
- 4) Refer to section 6.4.2

